

# Feature Engineering

Project: Responsible Plant Classifier

## 1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Learning Curves

Without opening the old lesson, explain one key idea from Learning Curves. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Reproducible Experiments

Without opening the old lesson, explain one key idea from Reproducible Experiments. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Hold Out Test Data

Without opening the old lesson, explain one key idea from Hold Out Test Data. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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## 2. Practise with fading help

Use W2: Transform validation for Feature engineering. Record the answer once; the lesson and workbook refer to the same attempt.

**Given inputs · Procedure I chose · Middle steps · Result and check**

**Given inputs:** \_\_\_\_\_

**Procedure I chose:** \_\_\_\_\_

**Middle steps:** \_\_\_\_\_

**Result and check:** \_\_\_\_\_

\_\_\_\_\_

**The step I completed without help, and the hint I needed**

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## 3. Individual reasoning check

Use the worked case for Feature engineering. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

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## 4. Independent transfer case

## Independent case: Apply independently

Training areas [4, 8, 12]. A new rectangle has length 3 width 4. Compute the training mean and centered new feature; state units.

### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## 5. Your learning target

Length=12 cm, width=30 mm. Compute length/width using matching units. Explain handling for zero or missing width and where learned scaling is fitted.

## 6. Solve the given case

Features are leaf length and width in cm; add length/width.

Compute for 8 and 2; state boundary checks.

Show your trace, calculation or evidence

### Write or draw your reasoning and evidence.

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## 7. Check a changed case

A feature is leaf\_length divided by leaf\_width. A row has width zero. What should the preprocessing rule do before division?

My independent answer and reason

**Write or draw your reasoning and evidence.**

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### Week 29 Test and reflect

## 8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

### Test record · My evidence

**Test record:** Changed input or condition

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Expected answer or fallback

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Observed result

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Why the result occurred

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

## 9. Project checkpoint

Add only justified Plant Classifier features, with formula, source fields, expected benefit, and leakage test.

The project change and evidence I will keep

**Write or draw your reasoning and evidence.**

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## 10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review \_\_\_\_\_ Next step \_\_\_\_\_

## 11. Transfer practice from the original programme

Length=12 cm, width=30 mm. Compute length/width using matching units. Explain handling for zero or missing width and where learned scaling is fitted.

**My independently reasoned answer**

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Record support used: none, hint or teacher explanation.

## 12. W1 Compute features

Compute the worked areas and their training mean.

**My reasoning and evidence**

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## 13. W2 Transform validation

Center the validation area using the frozen training mean.

**My reasoning and evidence**

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## 14. W3 Reject leakage

Why is delivery delay invalid for a prediction made before dispatch?

**My reasoning and evidence**

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## 15. W4 Apply independently

Training areas [4, 8, 12]. A new rectangle has length 3 width 4. Compute the training mean and centered new feature; state units.

**My reasoning and evidence**

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## 16. W5 Plan a fair comparison

How would you test adding area without claiming it must help?

**My reasoning and evidence**

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## 17. Relationships and confounding — W1 Describe the evidence

State the pattern in the three worked records without using caused or guaranteed.

**My reasoning and evidence**

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## 18. Relationships and confounding — W2 Identify a confounder

Explain how prior experience could relate to both practice time and score.

**My reasoning and evidence**

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## 19. Relationships and confounding — W3 Repair an overclaim

A learner predicts every student gains 2 points per extra hour. Explain why that is unsupported.

**My reasoning and evidence**

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## 20. Relationships and confounding — W4 Plan independently

A fictional dataset links more hints used with lower scores. Give a noncausal explanation and plan a fairer comparison of whether hints help.

**My reasoning and evidence**

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## 21. Relationships and confounding — W5 State a remaining limit

Even a better comparison has only four fictional groups. What should be reported?

**My reasoning and evidence**

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## 22. Feedback, return check and learning record

**Evidence · My record**

**Evidence:** Worked alone / hint / model

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** First step I repaired and why

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** New case checked later; date and result

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** What I can now explain without notes

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** One remaining question

**My record:** \_\_\_\_\_

\_\_\_\_\_

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

## 23. Pictorial case practice: Feature Engineering

Engineer a feature with a transformation defined before evaluating a new leaf.

Training rectangular leaf cards have areas [2, 6, 10] cm<sup>2</sup>. Area=length×width; center by training mean.  
Query length 2, width 4 cm.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

# Feature Engineering

## Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



### 1. Construct area

Query area =  $2 \times 4 = 8 \text{ cm}^2$ .



### 2. Fit centering

Training mean =  $(2 + 6 + 10) / 3 = 6 \text{ cm}^2$ .



### 3. Apply unchanged transform

Centered query feature =  $8 - 6 = 2 \text{ cm}^2$ .

## CHANGE ONE THING

Include the query in the centering mean.

Explain the changed result using the matching lesson.

What should be saved so another group transforms the query identically? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

### My picture-based explanation and changed-case prediction

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Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

## 24. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Prepare the demonstration. Gather reproducible instructions, evidence, access checks and remaining limitations.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

### Evidence to keep · My record

**Evidence to keep:** This week's input and deciding step

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence to keep:** Prediction and actual result (or labelled paper trace)

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence to keep:** One change since the previous checkpoint and why

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence to keep:** One error or limit and the next test

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

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## 25. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Engineer a feature with a transformation defined before evaluating a new leaf.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Centered query feature= $8-6=2\text{ cm}^2$ .

1. Construct area
2. Fit centering
3. Apply unchanged transform

#### My selected conclusion, deciding fact and explanation

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Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Apply unchanged transform Centered query feature= $8-6=2\text{ cm}^2$ . Read the picture in order. Centered query feature= $8-6=2\text{ cm}^2$ . The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

### Practice 2: explain the deciding evidence

Changed-case practice: Include the query in the centering mean. Which conclusion is supported by the supplied facts?

1. Mean becomes  $26/4=6.5\text{ cm}^2$  and query feature  $1.5\text{ cm}^2$ ; this changes the fitted transformation.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

### My selected conclusion, deciding fact and explanation

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Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Mean becomes  $26/4=6.5 \text{ cm}^2$  and query feature  $1.5 \text{ cm}^2$ ; this changes the fitted transformation. A held-out input should not silently redefine a training-fitted preprocessing step.

### Repair a missing building block

Review Learning Curves (ai-4/28). Goal: Read training-size learning curves. Short practice: What comparison is valid if the validation set changes between sizes? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Reproducible Experiments (ai-4/26). Goal: Record enough detail to reproduce a run. Short practice: What should accompany the mean 70%? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Hold Out Test Data (ai-4/23). Goal: Freeze the full evaluation procedure before testing. Short practice: What is needed for an honest final claim about the revised version? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

### Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

### What I repaired, the evidence I kept and the next test

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